

INFO 7250: Final Project

Toronto Traffic Analysis in Apache Hadoop

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Dataset: Toronto Traffic Volumes at Intersections for All Modes

This dataset can be found in the City of Toronto Open Data Portal here:

<https://open.toronto.ca/dataset/traffic-volumes-at-intersections-for-all-modes/>

The dataset contains the following files:

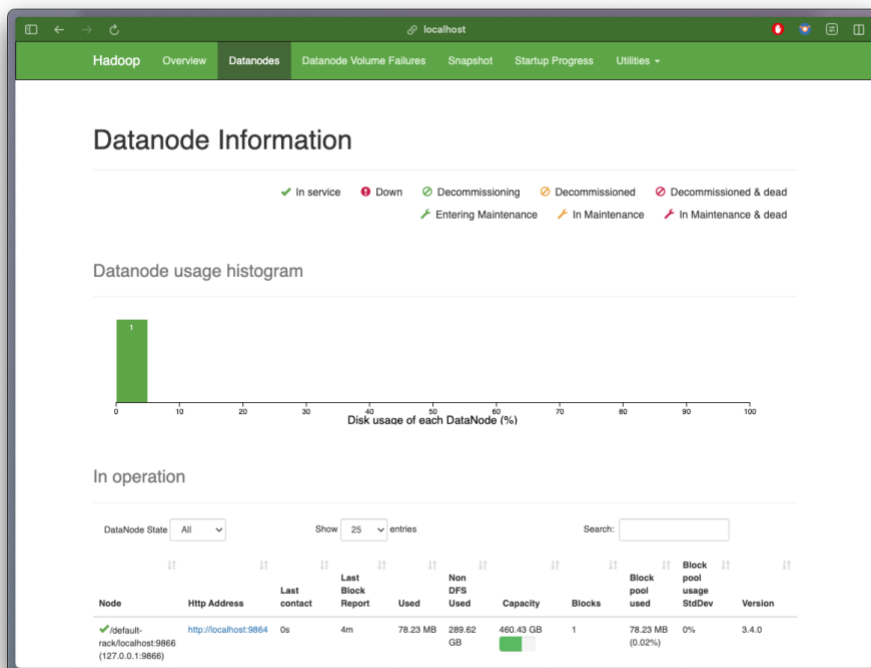
Name	Columns	Rows	Size
Toronto_Traffic_Volumes_2010-2019.csv	60	224,987	81.4 MB
Toronto_Traffic_Volumes_2020-2029.csv	60	193,157	49.7 MB
		418,144	131.1 MB

Starting HDFS

```
bin - zsh - 80x24
~ cd hadoop-3.4.0/sbin 4s base openjdk64-11.0.25
~/hadoop-3.4.0/sbin ./start-all.sh base openjdk64-11.0.25
WARNING: Attempting to start all Apache Hadoop daemons as rishabhkaushick in 10
seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [Rishabhs-MacBook-Pro.local]
2024-11-29 23:36:28,885 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Starting resourcemanager
Starting nodemanagers

~/hadoop-3.4.0/sbin jps 25s base openjdk64-11.0.25
5905 NodeManager
5362 NameNode
5609 SecondaryNameNode
5803 ResourceManager
6011 Jps
5471 DataNode

~/hadoop-3.4.0/sbin cd ../bin base openjdk64-11.0.25
```



Transferring the CSV Dataset to HDFS

```
bin — -zsh — 80x24
hdfs oom-listener

~/hadoop-3.4.0/bin > ./hadoop fs -ls /
2024-11-29 23:37:10,024 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Found 1 items
drwxr-xr-x - rishabhkaushick supergroup 0 2024-11-29 23:28 /toronto-t
raffic-project

~/hadoop-3.4.0/bin > ./hadoop fs -put ~/Documents/Northeastern/INF07250BigD
ata/Project/Toronto_Traffic_Volumes_2010-2019.csv /toronto-traffic-project
2024-11-29 23:37:29,899 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable

~/hadoop-3.4.0/bin > ./hadoop fs -put /Users/rishabhkaushick/Documents/Nort
heastern/INF07250BigData/Project/Toronto_Traffic_Volumes_2020-2029.csv /Users/ri
shabhkaushick/Documents/Northeastern/INF07250BigData/Project/Toronto_Traffic_Vol
umes_Intersection_Locations.csv /Users/rishabhkaushick/Documents/Northeastern/IN
F07250BigData/Project/Toronto_Traffic_Volumes_Metadata.csv /toronto-traffic-proj
ect
2024-11-29 23:42:38,088 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable

~/hadoop-3.4.0/bin > 
```

localhost

Hadoop Overview Datanodes Datanode Volume Failures Snapshot Startup Progress Utilities

Browse Directory

/toronto-traffic-project Go!

Show 25 entries Search:

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
-rw-r--r--	rishabhkaushick	supergroup	77.62 MB	Nov 29 23:37	1	128 MB	Toronto_Traffic_Volumes_2010-2019.csv
-rw-r--r--	rishabhkaushick	supergroup	47.41 MB	Nov 29 23:42	1	128 MB	Toronto_Traffic_Volumes_2020-2029.csv
-rw-r--r--	rishabhkaushick	supergroup	621 KB	Nov 29 23:42	1	128 MB	Toronto_Traffic_Volumes_Intersection_Locations.csv
-rw-r--r--	rishabhkaushick	supergroup	3.16 MB	Nov 29 23:42	1	128 MB	Toronto_Traffic_Volumes_Metadata.csv

Showing 1 to 4 of 4 entries Previous 1 Next

Hadoop, 2024.

Understanding the Data (in more detail)

Dataset: Toronto_Traffic_Volumes_2020-2029.csv

Sr. No	Column	Datatype	Description
1	_id	Integer	Primary Key column.
2	count_id	Integer	
3	count_date	Date	Date column.
4	location_id	Integer	Foreign key based on locations.
5	location	String	Name of the street intersection.
6	lng	Double	Longitude value.
7	lat	Double	Latitude value.
8	centreline_type	Integer	Not used in the project.
9	centreline_id	Integer	Not used in the project.
10	px	Integer	Not used in the project.
11	time_start	LocalDateTime	Data collection start time.
12	time_end	LocalDateTime	Data collection end time. Data was collected every 15 minutes – so end time is 15 minutes after the start time.x
13 to 60	sb_cars_r, sb_cars_t, sb_cars_l, nb_cars_r, nb_cars_t, nb_cars_l, wb_cars_r, wb_cars_t, wb_cars_l, eb_cars_r, eb_cars_t, eb_cars_l, sb_truck_r, sb_truck_t, sb_truck_l, nb_truck_r, nb_truck_t, nb_truck_l, wb_truck_r, wb_truck_t, wb_truck_l, eb_truck_r, eb_truck_t, eb_truck_l, sb_bus_r, sb_bus_t, sb_bus_l, nb_bus_r, nb_bus_t, nb_bus_l, wb_bus_r, wb_bus_t, wb_bus_l, eb_bus_r, eb_bus_t, eb_bus_l, sb_peds_r, sb_peds_t, sb_peds_l, nb_peds_r, nb_peds_t, nb_peds_l, wb_peds_r, wb_peds_t, wb_peds_l, eb_peds_r, eb_peds_t, eb_peds_l, nx_bike, sx_bike, ex_bike, wx_bike, nx_other, sx_other, ex_other, wx_other	Integer	Format: <direction_vehicle-type_lane> Directions: • sb: southbound • nb: northbound • eb: eastbound • wb: westbound Vehicle Types: • cars • trucks • buses • pedestrians • bikes • others Lane: • l: turning left • t: going straight (thru) • r: turning right

In this dataset, there are a total of **418,144 rows (193,157+224,987 rows)** and **60 columns**.

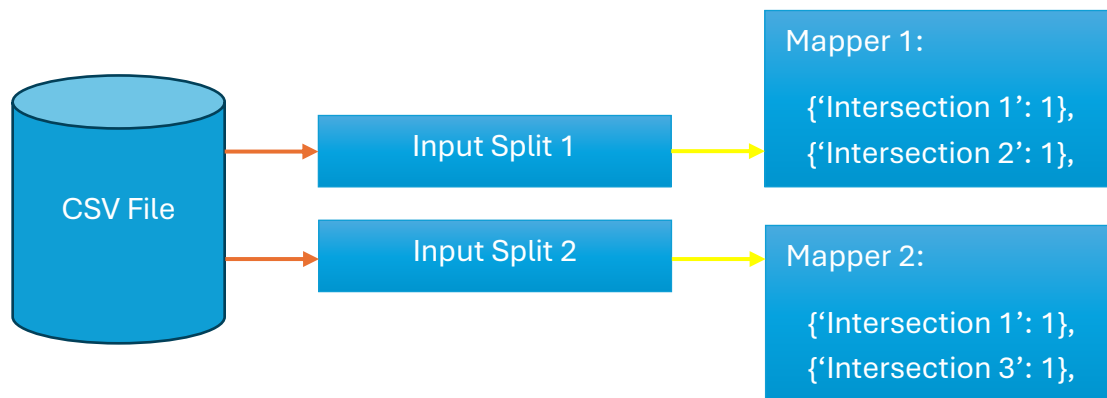
Map Reduce Algorithms:

Analysis #1: Counting Number of Unique Intersections

The idea is to understand the count of records for each intersection. This will also allow us to understand the total number of unique keys (location - street intersections).

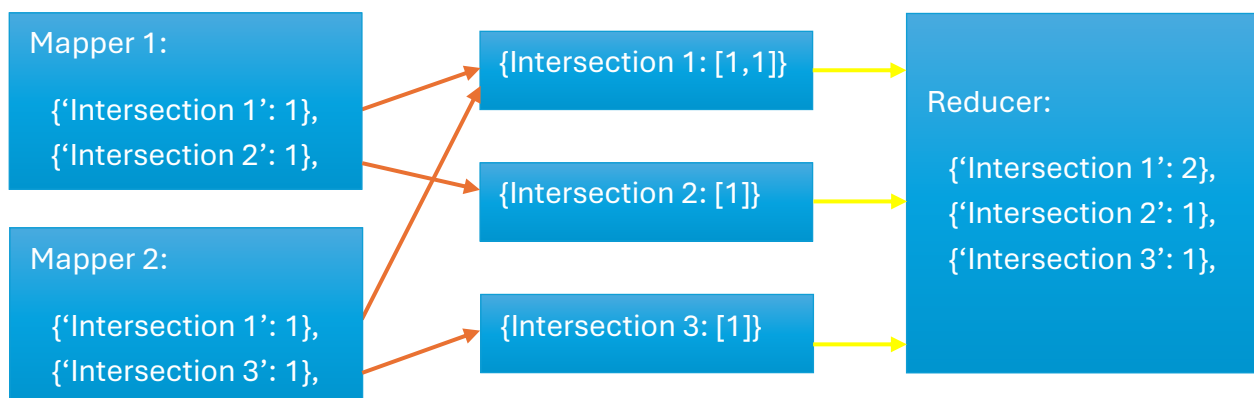
Intersection Count Mapper

This mapper will take the key as the 'location' and emit 1 as an *IntWritable* object for each row from the input-split.



Intersection Count Reducer

The reducer will get the inputs from the mapper, and it will sum all the numbers for each key and give the output as the key and sum of the count.



Combiner with Intersection Count Reducer

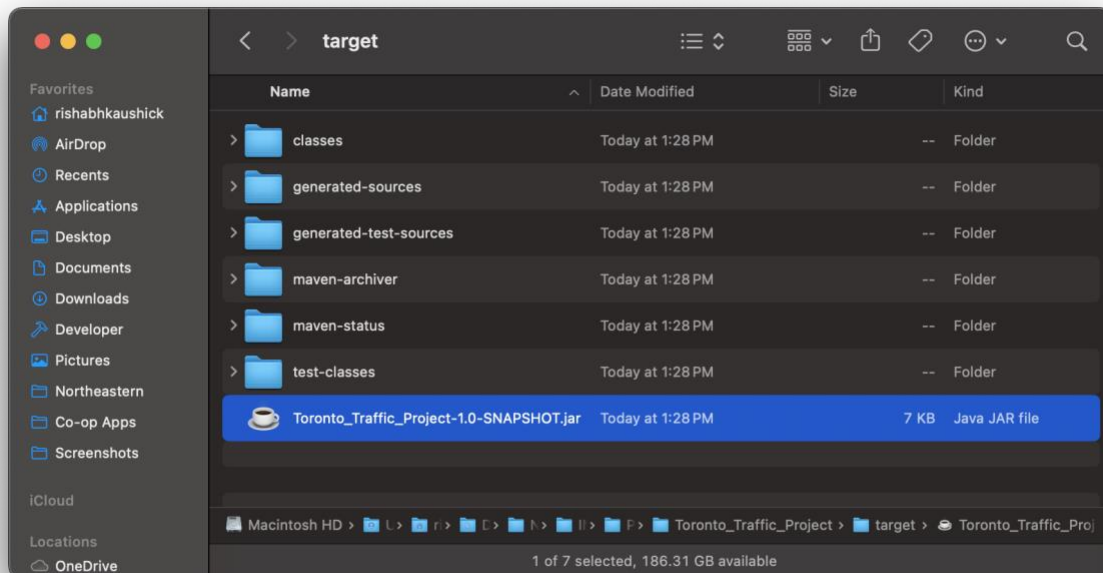
The operation performed in the reducer is both commutative and associative as it is just performing the sum. Therefore, we can use this reducer as a combiner to calculate the intermediate results.

For the implementation – please refer to the **Netbeans Project: Toronto_Traffic_Project**:

Class Type	Package	Class Name
Driver Class	com.mycompany.toronto_traffic_project	Toronto_Traffic_Project
Mapper Class	com.mycompany.mr	IntersectionCountMapper
Reducer Class	com.mycompany.mr	IntersectionCountReducer
Combiner Class	com.mycompany.mr	IntersectionCountReducer

Screenshots of the Map Reduce Execution

After building the project:



Running the map reduce on Hadoop with the command:

```
./hadoop jar
/Users/rishabhkaushick/Documents/Northeastern/INFO7250BigData/Project/Toronto_Traffic_Project/target/Toronto_Traffic_Project-1.0-SNAPSHOT.jar
com.mycompany.toronto_traffic_project.Toronto_Traffic_Project /toronto-traffic-project
/mr1_count_intersections
```

```
bin -- zsh -- 160x38
~/hadoop-3.4.0/bin > ./hadoop jar /Users/rishabhkaushick/Documents/Northeastern/INFO7250BigData/Project/Toronto_Traffic_Project/target/Toronto_Traffic_Project-1-SNAPSHOT.jar com.mycountry.toronto_traffic_project.Toronto_Traffic_Project /toronto-traffic-project /mr1_count_intersections
2024-12-10 15:54:09,085 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2024-12-10 15:54:10,321 INFO client.DefaultHARMPFallbackProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2024-12-10 15:54:10,663 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2024-12-10 15:54:10,680 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/rishabhkaushick/.staging/job_1732941391235_0005
2024-12-10 15:54:11,307 INFO input.FileInputFormat: Total input files to process : 1
2024-12-10 15:54:11,788 INFO mapreduce.JobSubmitter: number of splits:1
2024-12-10 15:54:11,919 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1732941391235_0005
2024-12-10 15:54:11,919 INFO mapreduce.JobSubmitter: Executing with tokens: []
2024-12-10 15:54:12,012 INFO conf.Configuration: resource-types.xml not found
2024-12-10 15:54:12,012 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2024-12-10 15:54:12,080 INFO impl.YarnClientImpl: Submitted application application_1732941391235_0005
2024-12-10 15:54:12,095 INFO mapreduce.Job: The url to track the job: http://localhost:8088/proxy/application_1732941391235_0005/
2024-12-10 15:54:12,095 INFO mapreduce.Job: Running job: job_1732941391235_0005
2024-12-10 15:54:16,186 INFO mapreduce.Job: Job job_1732941391235_0005 running in uber mode : false
2024-12-10 15:54:16,190 INFO mapreduce.Job: map 0% reduce 0%
2024-12-10 15:54:20,282 INFO mapreduce.Job: map 100% reduce 0%
2024-12-10 15:54:25,340 INFO mapreduce.Job: map 100% reduce 100%
2024-12-10 15:54:26,375 INFO mapreduce.Job: Job job_1732941391235_0005 completed successfully
2024-12-10 15:54:26,473 INFO mapreduce.Job: Counters: 50
File System Counters
  FILE: Number of bytes read=126477
  FILE: Number of bytes written=871021
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=49709104
  HDFS: Number of bytes written=117170
  HDFS: Number of read operations=8
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=1
  Launched reduce tasks=1
```

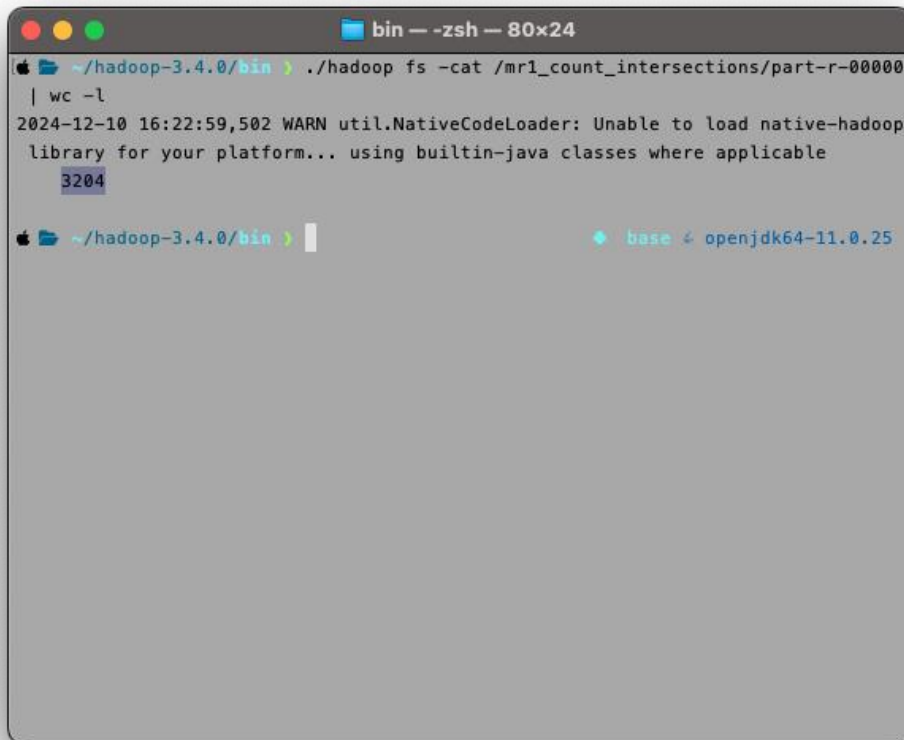
The Map Reduce operation took 17 seconds to execute. Getting the first few rows:

`./hadoop fs -head /mr1_count_intersections/part-r-00000`

```
bin -- zsh -- 80x24
~/hadoop-3.4.0/bin > ./hadoop fs -head /mr1_count_intersections/part-r-00000
0
.0.25
2024-12-10 15:55:24,958 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
# 2130 BAYVIEW AT VAUGHAN EST & 70M S OF VALLEYANNA DR (PX 1650) 32
# 4400 DUFFERIN ST AT DUFFERIN ST & 250M N STANSTEAD DR (PX 1625) 32
# 4600 BATHURST ST AT YMHA 335M S OF ELLERSLIE (PX 673) 32
# 475 THE WEST MALL DRIVEWAY 32
# 550 BAYVIEW AT 620M S OF POTTERY RD (PX 2194) 32
# 800 WARDEN AVE AT WARDEN AVE 1553M S OF CIVIC RD (PX 1638) 32
1630 LAWRENCE AVE W AT 55M E OF OLYMPIA DR (PX 1624) 32
ABELL ST AT QUEEN ST W 64
ABELL ST AT SUDBURY ST 32
ABERFOYLE CR AT BLOOR ST & GARDENVALE RD (PX 1402) 32
ABITIBI AVE AT WILLOWDALE AVE (PX 2164) 32
ACACIA RD AT DAVISVILLE AVE 32
ACACIA RD AT MILLWOOD RD N 32
ACACIA RD AT MILLWOOD RD S 32
ACORN AVE AT DUNDAS ST W (PX 2514) 32
ACORN AVE AT MONTESSON ST 32
ADELAIDE ST AT BAY ST (PX 62) 32
ADELAIDE ST AT BERKELEY ST (PX 1964) 64
ADELAIDE ST AT ONTARIO ST 96
```


To see the total number of unique intersections, we can run the following command:

```
./hadoop fs -cat /mr1_count_intersections/part-r-00000 | wc -l
```



```
bin — zsh — 80x24
~/hadoop-3.4.0/bin > ./hadoop fs -cat /mr1_count_intersections/part-r-00000 | wc -l
2024-12-10 16:22:59,502 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
3204
~/hadoop-3.4.0/bin > base ↵ openjdk64-11.0.25
```

We can see that there are a total of 3204 unique intersections. In the future when we are doing more map-reduce operations, we can use 3204 reducers for optimal performance.

Analysis #2: For Each Intersection Finding the Total Cars, Trucks, Busses & Pedestrians in every direction (nb, sb, eb, wb) in every lane (l, r, t)

Map Reduce in Java (Native Hadoop)

The idea is to take the 'Intersection Name' and calculate the total of the cars, busses, trucks etc in all directions and lanes.

For example: $\text{totalCars} = \text{sb_cars_r} + \text{sb_cars_t} + \text{sb_cars_l} +$

$\text{nb_cars_r} + \text{nb_cars_t} + \text{nb_cars_l} +$
 $\text{wb_cars_r} + \text{wb_cars_t} + \text{wb_cars_l} +$
 $\text{eb_cars_r} + \text{eb_cars_t} + \text{eb_cars_l}$

Class Type	Package	Class Name
Driver Class	com.mycompany.toronto_traffic_project	DriverClassTotalVehiclesAtIntersection
Mapper Class	com.mycompany.mr	TrafficMetricSumMapper
Reducer Class	com.mycompany.mr	TrafficMetricSumReducer
Combiner Class	com.mycompany.mr	TrafficMetricSumReducer
Partitioner	com.mycompany.mr	IntersectionPartitioner

Creating a Composite Key

First, we need to create a custom composite key. I have decided to take a generic composite key as follows:

Composite Key - <intersection, vehicleType>

Value: Total for the specific vehicle

Example: {Key: value}

- {<"Intersection1", "car">: 200}
- {<"Intersection2", "truck">: 15}

Custom Partitioner

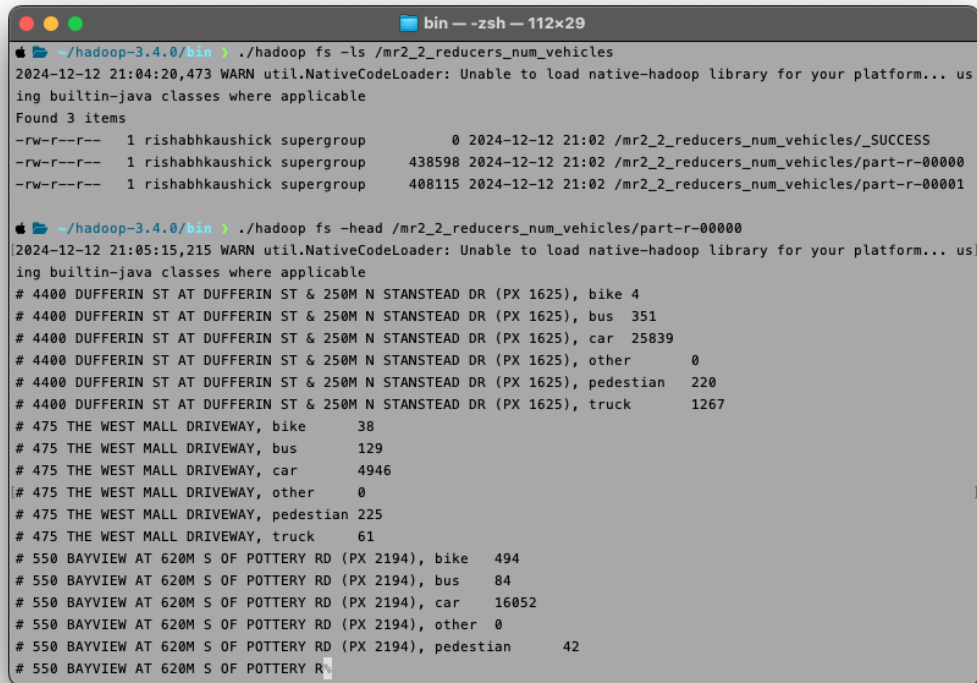
Since we want to create 3204 reducers, we need a custom partitioner to send the records to all the reducers.

With just 1 reducer it took 22 seconds.

With 2 reducers also it took 22 seconds.

With 3204 reducers – it kept running for 30 minutes and only reached 23%. It was too many reducers for this task.

Screenshot with 2 Reducers:



```
bin -- zsh -- 112x29
~/hadoop-3.4.0/bin ./hadoop fs -ls /mr2_2_reducers_num_vehicles
2024-12-12 21:04:20,473 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 3 items
-rw-r--r-- 1 rishabhkaushick supergroup 0 2024-12-12 21:02 /mr2_2_reducers_num_vehicles/_SUCCESS
-rw-r--r-- 1 rishabhkaushick supergroup 438598 2024-12-12 21:02 /mr2_2_reducers_num_vehicles/part-r-00000
-rw-r--r-- 1 rishabhkaushick supergroup 408115 2024-12-12 21:02 /mr2_2_reducers_num_vehicles/part-r-00001

~/hadoop-3.4.0/bin ./hadoop fs -head /mr2_2_reducers_num_vehicles/part-r-00000
2024-12-12 21:05:15,215 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
# 4400 DUFFERIN ST AT DUFFERIN ST & 250M N STANSTEAD DR (PX 1625), bike 4
# 4400 DUFFERIN ST AT DUFFERIN ST & 250M N STANSTEAD DR (PX 1625), bus 351
# 4400 DUFFERIN ST AT DUFFERIN ST & 250M N STANSTEAD DR (PX 1625), car 25839
# 4400 DUFFERIN ST AT DUFFERIN ST & 250M N STANSTEAD DR (PX 1625), other 0
# 4400 DUFFERIN ST AT DUFFERIN ST & 250M N STANSTEAD DR (PX 1625), pedestrian 220
# 4400 DUFFERIN ST AT DUFFERIN ST & 250M N STANSTEAD DR (PX 1625), truck 1267
# 475 THE WEST MALL DRIVEWAY, bike 38
# 475 THE WEST MALL DRIVEWAY, bus 129
# 475 THE WEST MALL DRIVEWAY, car 4946
# 475 THE WEST MALL DRIVEWAY, other 0
# 475 THE WEST MALL DRIVEWAY, pedestrian 225
# 475 THE WEST MALL DRIVEWAY, truck 61
# 550 BAYVIEW AT 620M S OF POTTERY RD (PX 2194), bike 494
# 550 BAYVIEW AT 620M S OF POTTERY RD (PX 2194), bus 84
# 550 BAYVIEW AT 620M S OF POTTERY RD (PX 2194), car 16052
# 550 BAYVIEW AT 620M S OF POTTERY RD (PX 2194), other 0
# 550 BAYVIEW AT 620M S OF POTTERY RD (PX 2194), pedestrian 42
# 550 BAYVIEW AT 620M S OF POTTERY RD
```

24 seconds for 5 reducers → performance is starting to get worse.

```
bin --zsh -- 112x29

Input split bytes=148
Combine input records=1158948
Combine output records=19224
Reduce input groups=19224
Reduce shuffle bytes=909444
Reduce input records=19224
Reduce output records=19224
Spilled Records=38448
Shuffled Maps =5
Failed Shuffles=0
Merged Map outputs=5
GC time elapsed (ms)=88
CPU time spent (ms)=0
Physical memory (bytes) snapshot=0
Virtual memory (bytes) snapshot=0
Total committed heap usage (bytes)=1610612736

Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=49708956
File Output Format Counters
  Bytes Written=846713

~/hadoop-3.4.0/bin
```

```
bin --zsh -- 112x29

~/hadoop-3.4.0/bin ./hadoop fs -ls /mr2_5_reducers_num_vehicles
2024-12-12 21:16:31,550 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... us
ing builtin-java classes where applicable
Found 6 items
-rw-r--r-- 1 rishabhkaushick supergroup 0 2024-12-12 21:14 /mr2_5_reducers_num_vehicles/_SUCCESS
-rw-r--r-- 1 rishabhkaushick supergroup 162145 2024-12-12 21:14 /mr2_5_reducers_num_vehicles/part-r-00000
-rw-r--r-- 1 rishabhkaushick supergroup 176593 2024-12-12 21:14 /mr2_5_reducers_num_vehicles/part-r-00001
-rw-r--r-- 1 rishabhkaushick supergroup 176041 2024-12-12 21:14 /mr2_5_reducers_num_vehicles/part-r-00002
-rw-r--r-- 1 rishabhkaushick supergroup 165535 2024-12-12 21:14 /mr2_5_reducers_num_vehicles/part-r-00003
-rw-r--r-- 1 rishabhkaushick supergroup 166399 2024-12-12 21:14 /mr2_5_reducers_num_vehicles/part-r-00004

~/hadoop-3.4.0/bin ./hadoop fs -head /mr2_5_reducers_num_vehicles/part-r-00004
2024-12-12 21:16:47,051 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... us
ing builtin-java classes where applicable
ABELL ST AT QUEEN ST W, bike 1121
ABELL ST AT QUEEN ST W, bus 228
ABELL ST AT QUEEN ST W, car 12664
ABELL ST AT QUEEN ST W, other 0
ABELL ST AT QUEEN ST W, pedestrian 4600
ABELL ST AT QUEEN ST W, truck 452
ACORN AVE AT MONTESSON ST, bike 4
ACORN AVE AT MONTESSON ST, bus 3
ACORN AVE AT MONTESSON ST, car 460
ACORN AVE AT MONTESSON ST, other 0
ACORN AVE AT MONTESSON ST, pedestrian 26
ACORN AVE AT MONTESSON ST, truck 0
ADELAIDE ST AT ONTARIO ST, bike 255
ADELAIDE ST AT ONTARIO ST, bus 70
ADELAIDE ST AT ONTARIO ST, car 24441
```

Analysis #3: Filtering Design Pattern – Top 10

Class Type	Package	Class Name
Driver Class	com.mycompany.toronto_traffic_project	DriverClassTotalVehiclesAtIntersection
Mapper Class	com.mycompany.mr	Top10IntersectionsCarsMapper
Reducer Class	com.mycompany.mr	Top10IntersectionCarsReducer

Top 10 Intersections with Highest Number of Cars

```
bin -- zsh -- 166x35

Spilled Records=20
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=8
CPU time spent (ms)=0
Physical memory (bytes) snapshot=0
Virtual memory (bytes) snapshot=0
Total committed heap usage (bytes)=536870912

Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=1415175
File Output Format Counters
  Bytes Written=440

Top 10 Intersections with Highest Cars are:
AVENUE RD AT BLOOR ST W & QUEENS PARK 398313
MOUNT PLEASANT RD AT ST CLAIR AVE E 368428
YONGE ST AT CHAPLIN CRES & DAVISVILLE AVE (PX 47) 354084
DAVISVILLE AVE AT MOUNT PLEASANT RD 349288
ST CLAIR AVE AT YONGE ST 340942
AVENUE RD AT CHAPLIN CRES (PX 116) 306379
DANFORTH AVE AT VICTORIA PARK AVE 283947
DUNDAS ST W AT KEELE ST (PX 188) 275378
CHAPLIN CRES AT ORIOLE PKWY (PX 115) 245601
BLOOR ST W AT KEELE ST & PARKSIDE DR (PX 328) 233763

~/hadoop-3.4.0/bin 44s base openjdk64-11.0.25
```

Top 10 Intersections with Highest Number of Trucks

```
bin -- -zsh -- 166x35

Spilled Records=20
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=8
CPU time spent (ms)=0
Physical memory (bytes) snapshot=0
Virtual memory (bytes) snapshot=0
Total committed heap usage (bytes)=536870912

Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=1415175
File Output Format Counters
  Bytes Written=440

Top 10 Intersections with Highest Cars are:
AVENUE RD AT BLOOR ST W & QUEENS PARK    398313
MOUNT PLEASANT RD AT ST CLAIR AVE E      368428
YONGE ST AT CHAPLIN CRES & DAVISVILLE AVE (PX 47)  354084
DAVISVILLE AVE AT MOUNT PLEASANT RD    349288
ST CLAIR AVE AT YONGE ST                348942
AVENUE RD AT CHAPLIN CRES (PX 116)      306379
DANFORTH AVE AT VICTORIA PARK AVE       283947
DUNDAS ST W AT KEELE ST (PX 188)       275378
CHAPLIN CRES AT ORIOLE PKWY (PX 115)    245601
BLOOR ST W AT KEELE ST & PARKSIDE DR (PX 328)  233763

~/hadoop-3.4.0/bin > | 44s base + openjdk64-11.0.25
```

Top 10 Intersections with Highest Number of Pedestrians

```
bin -- -zsh -- 166x35

Reduce input records=10
Reduce output records=10
Spilled Records=20
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=9
CPU time spent (ms)=0
Physical memory (bytes) snapshot=0
Virtual memory (bytes) snapshot=0
Total committed heap usage (bytes)=536870912

Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=1415175
File Output Format Counters
  Bytes Written=389

Top 10 Intersections with Highest Pedestrians are:
BLOOR ST AT YONGE ST    369215
ST CLAIR AVE AT YONGE ST  322084
YONGE ST AT CHAPLIN CRES & DAVISVILLE AVE (PX 47)  244843
AVENUE RD AT BLOOR ST W & QUEENS PARK    218035
BAY ST AT QUEEN ST W    151281
YONGE ST AT CARLTON ST & COLLEGE ST    120018
DUNDAS ST W AT SPADINA AVE    117716
YONGE ST AT CHURCH ST & DAVENPORT RD (PX 41)  117494
BAY ST AT BLOOR ST W    115769
BLOOR ST W AT DUFFERIN ST (PX 325)  107952

~/hadoop-3.4.0/bin > | 42s base + openjdk64-11.0.25
```

Conclusion

This project had been able to find the busiest street intersections for the city of Toronto. Sharing this analysis with the City of Toronto could help city planners have a better understanding of the busy city intersections to help build more roads or public transport to divert traffic from these busy areas.

Future Scope

After this course, I would like to continue to perform Map Reduce analysis for the following:

- Analyzing the direction of the flow of traffic (northbound, southbound, eastbound, westbound) for each type of vehicle and all types of vehicles.
- For each location, calculating total vehicle count per year to understand how the traffic has changed over time.